

Assignment 7

Encapsulation with Class and Objects

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Objective

The objective of this assignment is to implement a simple Bank Account Management System in Java using the concept of Encapsulation, by binding data (instance variables) and methods together within a single class, restricting direct access to the data, and providing controlled access through validated getter and setter methods.

Concept: Encapsulation

Encapsulation refers to the technique of wrapping data (instance variables) and the methods that operate on that data into a single unit, i.e., a class, while restricting direct access to the internal state of an object from outside the class.

Problem: Bank Account Management System

```
public class BankAccount {

    // Private data members (Encapsulation)
    private String accountNumber;
    private String accountHolderName;
    private double balance;

    // Parameterized constructor with validation
    public BankAccount(String accountNumber, String accountHolderName, double balance)
    {
        setAccountNumber(accountNumber);
        setAccountHolderName(accountHolderName);
        setBalance(balance);
    }

    // Getter for Account Number
    public String getAccountNumber() {
        return accountNumber;
    }

    // Setter for Account Number with validation
    public void setAccountNumber(String accountNumber) {
        if (accountNumber == null || accountNumber.trim().isEmpty()) {
            System.out.println("Invalid Account Number: cannot be empty");
        } else if (accountNumber.length() < 5) {
            System.out.println("Invalid Account Number: must contain at least 5
characters");
        } else {
            this.accountNumber = accountNumber;
        }
    }

    // Getter for Account Holder Name
    public String getAccountHolderName() {
        return accountHolderName;
    }

    // Setter for Account Holder Name with validation
    public void setAccountHolderName(String accountHolderName) {
        if (accountHolderName == null || accountHolderName.trim().isEmpty()) {
```

```

        System.out.println("Invalid Name: cannot be empty");
    } else if (accountHolderName.length() < 3) {
        System.out.println("Invalid Name: must contain at least 3 characters");
    } else {
        this.accountHolderName = accountHolderName;
    }
}

// Getter for Balance
public double getBalance() {
    return balance;
}

// Setter for Balance with validation
public void setBalance(double balance) {
    if (balance < 0) {
        System.out.println("Invalid Balance: cannot be negative");
    } else {
        this.balance = balance;
    }
}

// Method to deposit amount
public void deposit(double amount) {
    if (amount > 0) {
        balance = balance + amount;
    } else {
        System.out.println("Invalid Deposit Amount");
    }
}

// Method to withdraw amount
public void withdraw(double amount) {
    if (amount > 0 && amount <= balance) {
        balance = balance - amount;
    } else if (amount > balance) {
        System.out.println("Insufficient Balance");
    } else {
        System.out.println("Invalid Withdrawal Amount");
    }
}

// Method to display account details
public void displayAccountDetails() {
    System.out.println("Account Number: " + accountNumber);
    System.out.println("Account Holder Name: " + accountHolderName);
    System.out.println("Available Balance: " + balance);
}

public static void main(String[] args) {
    // Creating a BankAccount object using the parameterized constructor
    BankAccount account = new BankAccount("ACC1001", "Rahul Sharma", 5000.0);

    // Display initial account details
    account.displayAccountDetails();

    // Depositing amount

```

```
        System.out.println("\nDepositing: 2000");
        account.deposit(2000);

        // Withdrawing amount
        System.out.println("Withdrawal: 3000");
        account.withdraw(3000);

        // Display updated balance
        System.out.println("\nUpdated Balance: " + account.getBalance());
    }
}
```

Output

```
Account Number: ACC1001
Account Holder Name: Rahul Sharma
Available Balance: 5000.0

Depositing: 2000
Withdrawal: 3000

Updated Balance: 4000.0
```